

TAO LIN

State Key Lab of CAD&CG, Zhejiang University ◇ 310058 ◇ Hangzhou, China
http://nblintao.github.io/ ◇ (86) 15957445452 ◇ nblintao@gmail.com

EDUCATION

Zhejiang University

B.Eng. in Computer Science and Technology

Oct. 2012 - Present

Hangzhou, China

Overall GPA: 3.96 / 4.00 (92.45 / 100)

The third year GPA: 4.00 / 4.00 (94.43 / 100)

Rank: 1st in Qizhen Honors Class, Chu Kochen Honors College (Top 2%)

Hong Kong University of Science and Technology (HKUST)

Exchange Student for Final Year Project

Sept. 2015 - Feb. 2016

Hong Kong

PUBLICATIONS

Mobility Viewer: A Eulerian Approach for Studying Urban Crowd Flow

Yuxin Ma, **Tao Lin**, Zhendong Cao, Chen Li, Wei Chen

IEEE Transactions on Intelligent Transportation Systems. Accepted, 2015. [\[PDF\]](#) [\[Video\]](#)

TieVis: Visual Analytics of Evolution of Interpersonal Ties

Tao Lin, Fangzhou Guo, Yingcai Wu, Wei Chen, Biao Zhu, Sidong Wang, Huamin Qu

IDAVis 2016: Workshop on Intelligent Data Analytics and Visualization. Submitted. [\[PDF\]](#) [\[Video\]](#) [\[Code\]](#)

EXPERIENCE

Visual Analytics Group, State Key Lab of CAD&CG

Research Assistant

June 2013 - Aug. 2015

Zhejiang University, China

Research Advisor: Professor Wei Chen [\[Homepage\]](#)

- Administrated a computer cluster with 20 nodes for more than one year.
- Processed large-scale data on the group's cluster with Hadoop and Spark for more than one year.
- **Project Mobility Viewer** - A visual analytics system supporting situation-aware understanding and visual reasoning of human mobility. [\[PDF\]](#) [\[Video\]](#)
Core Team Member
 - Preprocessed tera-scale data on the cluster with Spark.
 - Designed a distributed algorithm to detect and eliminate ping-pong effects.
 - Designed and implemented the Flow Volume Analysis Subsystem.
 - Technique: MapReduce, Hadoop, Spark, Python, JavaScript (D3, WebGL), OpenStreetMap.
- **Project TieVis** - A visual analytics framework and system that enables interactive analysis and exploration of the dynamic changes of interpersonal ties. [\[PDF\]](#) [\[Video\]](#) [\[Code\]](#)
Project Leader
 - Preprocessed the data with Python (Extract, Aggregate, PCA and MDS).
 - Designed and implemented the whole TieVis System.
 - Technique: Python(NumPy, SciPy, scikit-learn), JavaScript.

Multimedia Technology Research Center

Visiting Intern

Sept. 2015 - Feb. 2016

HKUST, Hong Kong

Supervisor: Professor S.-H. Gary Chan [\[Homepage\]](#)

- Worked with a team that has been developing Streamphony, a next-generation streaming cloud for large-scale high bit-rate applications over the global Internet; helped turn research results into a practical system for commercial trials.
- Implemented the forward error correction (FEC) module with Reed-Solomon codes.
- Improved the quality and robustness of the video stream.
- Currently developing a data analysis system for the proxy records with HBase.
- Technique: C++ (STL, Boost (asynchronous, memory pool and smart pointer)), lock-free programming, code review, unit testing (Google Test, Google Mock), Hadoop (HBase).

HONORS AND AWARDS

Google Excellence Scholarship <i>One of 58 bachelor and master's students in Mainland China</i>	2015 Google Inc.
National Scholarship <i>Top 1% students</i>	2014 - 2015 Ministry of Education, China
First-Class Scholarship for Outstanding Students <i>Top 3% students</i>	2013 - 2014, 2014 - 2015 Zhejiang University
Honorable Mention, Interdisciplinary Contest in Modeling	2015
Second Prize, Zhejiang University Programming Contest	2014
Third Prize, ACM-ICPC China Zhejiang Provincial Programming Contest	2014

SELECTED PROJECTS

MiniSQL Database Management System <i>Team Leader</i>	Sept. 2014 - Nov. 2014 <i>Course Project</i>
· Designed and implemented a DBMS, which can create/delete/query tables/indexes/records by interpreting SQL. · Modules: Interpreter, API, Catalog Manager, Record Manager, Index Manager, Buffer Manager. · Technique: C++, Parser, Cache Replacement (LRU), B+ Tree.	
Visual Analysis of Library Data <i>Team Leader</i> [Video] [Code]	Apr. 2014 - May 2015 <i>Student Research Training Program</i>
· Promoted library users' decision-making by interacting with real records. · Designed and implemented a visual analysis system. · Technique: JavaScript (D3, Bootstrap, ECharts and jQuery), Python (Django), Oracle Database.	
Precomputed Shadow Fields for Dynamic Scenes <i>Independent</i> [Code]	Mar. 2015 - July 2015 <i>Course Project</i>
· Implemented a system for rendering dynamic soft shadows based on the methodology outlined in a paper from SIGGRAPH 2005. · Course: Advances in Computer Graphics, Score: 94/100, Instructor: Kun Zhou. · Technique: C++ (DirectX), HLSL, precomputed radius transfer, spherical harmonics.	
A 3-D Auto-Stereoscopic Front Projection System <i>Team Member</i> [Video]	July 2014 - May 2015 <i>Challenge Cup Competition</i>
· Worked with a team of optical engineering students to achieve glasses-free 3D by using 60 projectors. · Derived the core mathematical formulas by light field reconstruction principles. · Implemented the prototype with OpenGL.	

TECHNICAL STRENGTHS

Computer Languages	C/C++, JavaScript, Python, Java, MATLAB, Verilog
Libraries & Frameworks	Visualization (D3, ECharts) C++ unit testing (Google Test, Google Mock)
Tools	Computer graphics & vision (OpenGL, DirectX, WebGL, OpenCV) Data science (Hadoop, Spark, Hive, HBase, MySQL) Software engineering (Unix Command, Makefile, Git, Phabricator)